



Industrial Machine Learning on Hadoop and Spark

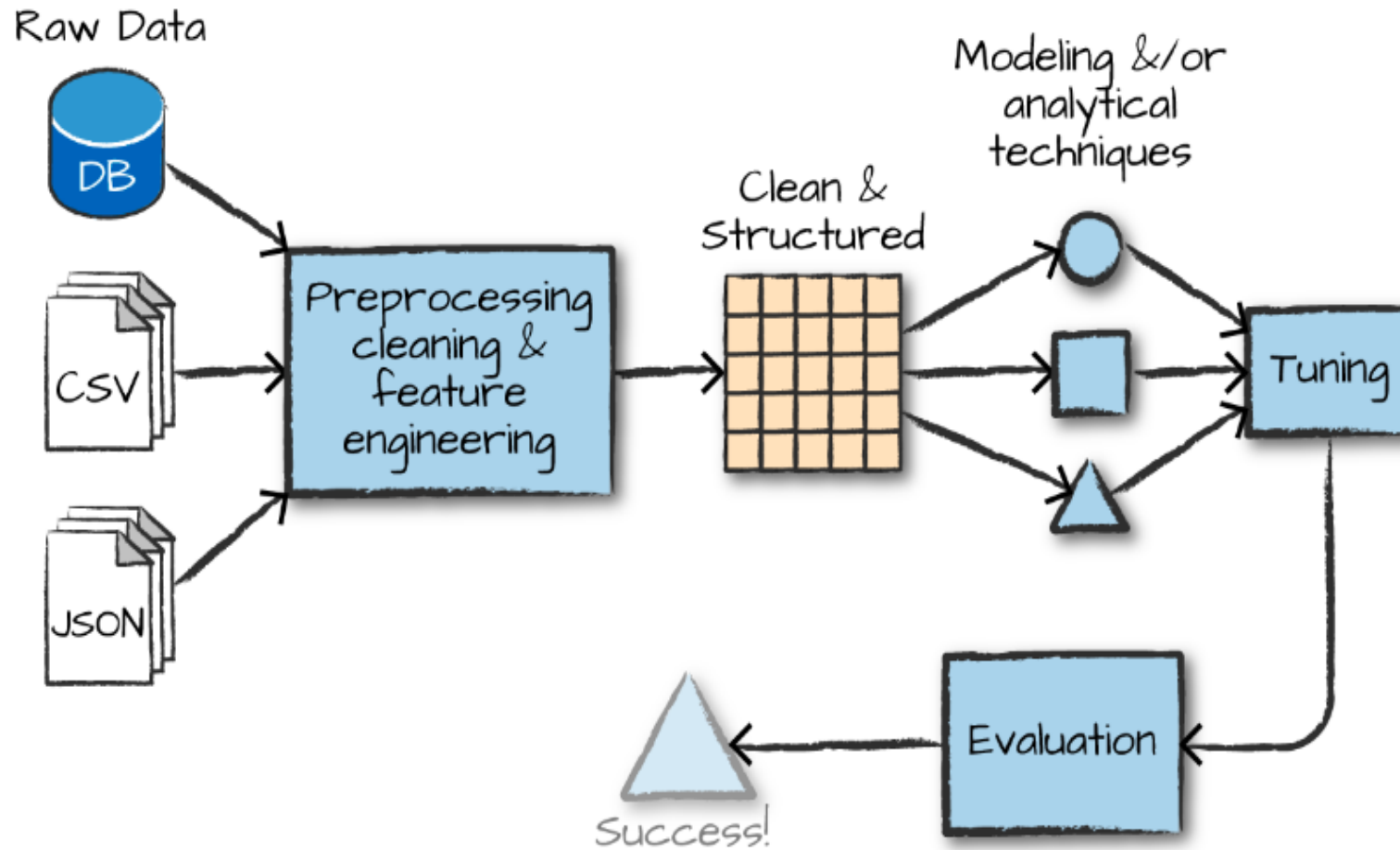
Maks Nakhodnov, Bremen 2025

Distributed computing. Spark

Plan:

- Spark ML

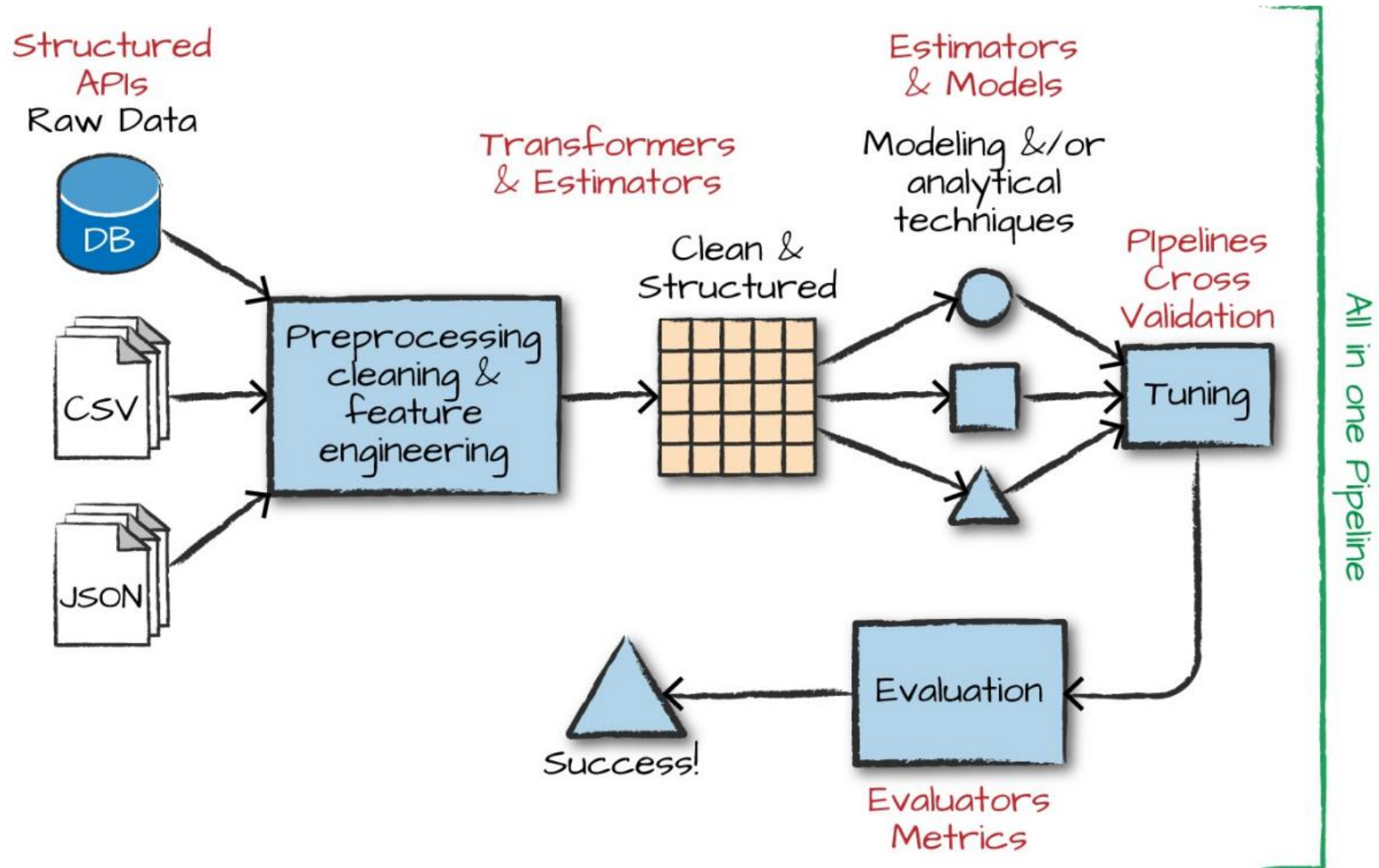
What do we want to do?



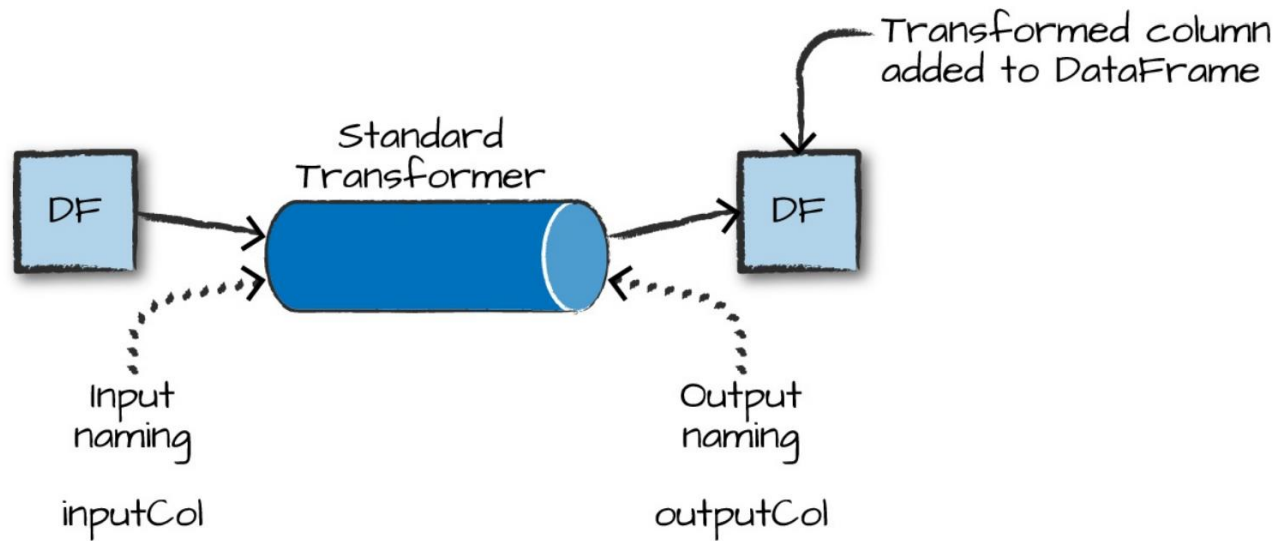
What does Spark ML allow you to do?

<http://spark.apache.org/docs/latest/ml-features.html>

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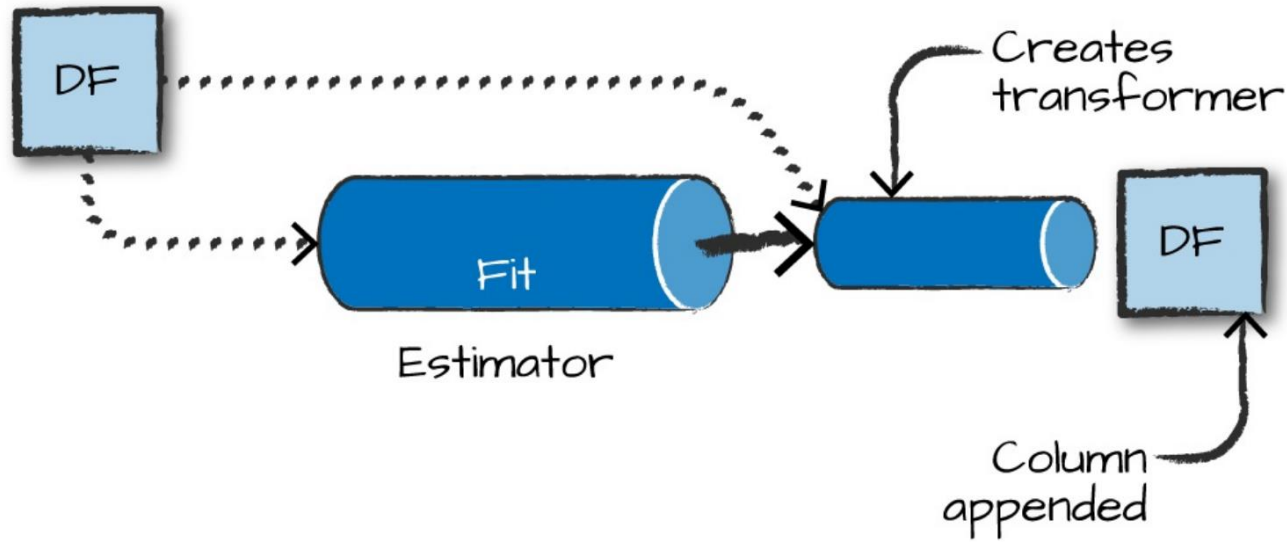


Spark ML. Transformer



- Tokenizer
- StopWordsRemover
- n-gram
- Binarizer
- PCA
- PolynomialExpansion
- Discrete Cosine Transform (DCT)
- StringIndexer
- IndexToString
- OneHotEncoder
- VectorIndexer
- Interaction
- Normalizer
- StandardScaler
- RobustScaler
- MinMaxScaler
- MaxAbsScaler
- Bucketizer
- ElementwiseProduct
- SQLTransformer
- VectorAssembler
- VectorSizeHint
- QuantileDiscretizer
- Imputer

Spark ML. Estimator



- [KMeans](#)
- [LDA](#)
- [GaussianMixture](#)
- [ALS](#)
- [DecisionTreeRegressor](#)
- [LinearRegression](#)
- [RandomForestRegressor](#)
- [LogisticRegression](#)
- [DecisionTreeClassifier](#)
- [RandomForestClassifier](#)
- [NaiveBayes](#)
- [OneVsRest](#)
- [MultilayerPerceptronClassifier](#)

Spark ML. Evaluator

```
class Evaluator(Params, metaclass=ABCMeta):
    """
    Base class for evaluators that compute metrics from predictions.
    """

    def evaluate(self, dataset: DataFrame, params: Optional["ParamMap"] = None) -> float:
        """
        Evaluates the output with optional parameters.

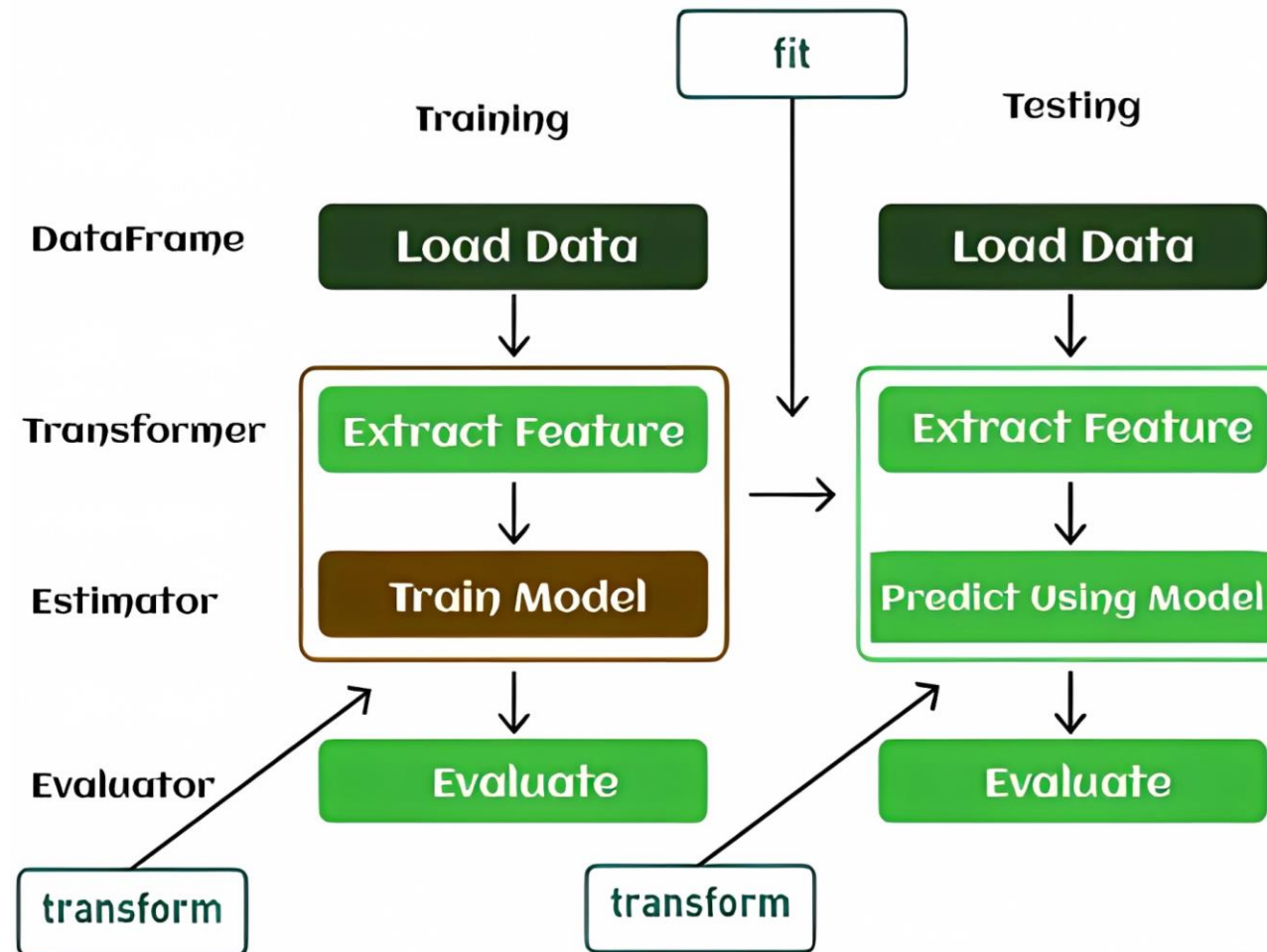
        Parameters
        -----
        dataset : :py:class:`pyspark.sql.DataFrame`
            a dataset that contains labels/observations and predictions
        params : dict, optional
            an optional param map that overrides embedded params

        Returns
        -----
        float
            metric
        """
        pass

    def isLargerBetter(self) -> bool:
        """
        Indicates whether the metric returned by :py:meth:`evaluate` should be maximized
        (True, default) or minimized (False).
        A given evaluator may support multiple metrics which may be maximized or minimized.
        """
        pass
```

- [BinaryClassificationEvaluator](#)
- [RegressionEvaluator](#)
- [MulticlassClassificationEvaluator](#)
- [MultilabelClassificationEvaluator](#)
- [ClusteringEvaluator](#)

Spark ML Workflow



Spark ML Pipeline

